

Modelling & Simulation: Assignment 4 (option 2)

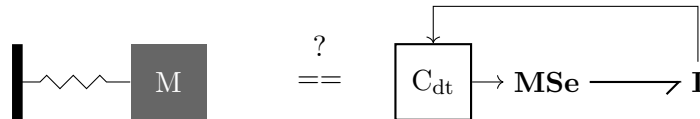
Measuring and controlling energy flows

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Introduction

Impedance control is a well-established framework for shaping physical interactions of robotic systems with their environment. In its most basic form, it relies on mimicking physical spring-dampers through controlled actuation. While this can be shown to be stable and passive in its definition, proving such properties in practical digital control is not trivial, due to discrete-time, delays, and other practical implementation issues.

In general, literature does not explicitly deal with discrete time control implementations and their performance implications, even though it cannot be guaranteed that a stable continuous time control system remains stable when it is implemented in a discrete time computer. In this assignment you will analyse such issues on a basic use case, and implement approaches to guarantee passivity, and thus, stability.



Suggested material:

- G.A. Folkertsma and S. Stramigioli, “Energy in Robotics”, Foundations and Trends in Robotics, vol. 6, no. 3, 2017.
- G.A. Folkertsma, S.S. Groothuis, and S. Stramigioli, “Safety and Guaranteed Stability through Embedded Energy-Aware Actuators”, IEEE International Conference on Robotics and Automation (ICRA), 2018.

See the next page for the assignment.

Assignment

Model and simulate a spring–mass system, first with a physical spring, and then using an electric motor¹ and discrete-time controller. Analyse the passivity of the systems, and passivate the discrete-time system.

1. Model a mass with a physical spring, and a mass with an equivalent impedance controller implemented in discrete-time. Compare the passivity and stability of the two systems, under various conditions (sample rates, system parameters). Show the various power flows in your systems.
2. Implement “passive energy sampling” for your impedance controller and show whether it accurately measures the true energy injected into the system.
3. *Passivate* the system and show that it is passive for all possible inertia values and spring constants, i.e. for all plant models.

Additional questions you could try to answer are:

- How does control delay (e.g. due to communication) affect your system? What about quantisation errors? Why?
- What if the motor is not a pure torque source? If you include the motor’s electrical dynamics and apply a voltage as input, can you still guarantee passivity? Why (not)?
- What physical properties of the system affect the passivity of the overall controlled system?

¹You can neglect the motor’s electrical dynamics and inertia, making it a pure torque source.